

Role of Anthocyanin-enriched Purple-fleshed Sweet Potato P40 in Colorectal Cancer Prevention

.

Scope Anthocyanins, the natural pigments in plant foods, have been associated with cancer prevention. However, the content of anthocyanins in staple foods is typically low and the mechanisms by which they exert anti-cancer activity is not yet fully defined. Methods and results We selected an anthocyanin-enriched purple-fleshed sweet potato clone, P40, and investigated its potential anti-cancer effect in both in vitro cell culture and in vivo animal model. In addition to a high level of total phenolics and antioxidant capacity, P40 possesses a high content of anthocyanins at 7.5 mg/g dry matter. Treatment of human colonic SW480 cancer cells with P40 anthocyanin extracts at 0–40 μ M of peonidin-3-glucoside equivalent resulted in a dose-dependent decrease in cell number due to cytostatic arrest of cell cycle at G1 phase but not cytotoxicity. Furthermore, dietary P40 at 10–30% significantly suppressed azoxymethane-induced formation of aberrant crypt foci in the colons of CF-1 mice in conjunction with, at least in part, a lesser proliferative PCNA and a greater apoptotic caspase-3 expression in the colon mucosal epithelial cells. Conclusion These observations, coupled with both in vitro and in vivo studies reported here, suggest anthocyanin-enriched sweet potato P40 may protect against colorectal cancer by inducing cell cycle arrest, anti-proliferative and apoptotic mechanisms.